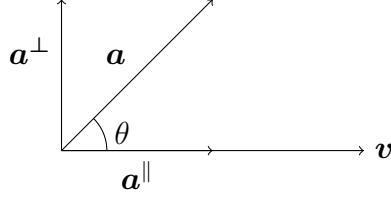


A. $\kappa = \frac{\mathbf{v} \times \mathbf{a}}{|\mathbf{v}|^3}$.

Proof. Consider the following setup



From the above picture, we can see that $\sin(\theta) = \frac{|\mathbf{a}^\perp|}{|\mathbf{a}|}$, which implies that

$$|\mathbf{a}| \sin(\theta) = |\mathbf{a}^\perp|.$$

Note that $|\mathbf{v} \times \mathbf{a}| = |\mathbf{v}| |\mathbf{a}| \sin(\theta)$. Using the equation from above, we see that $|\mathbf{v} \times \mathbf{a}| = |\mathbf{v}| |\mathbf{a}^\perp|$. Dividing by $|\mathbf{v}|^3$, we compute

$$\frac{|\mathbf{v} \times \mathbf{a}|}{|\mathbf{v}|^3} = \frac{|\mathbf{v}| |\mathbf{a}^\perp|}{|\mathbf{v}|^3} = \frac{|\mathbf{a}^\perp|}{|\mathbf{v}|^2} = \kappa.$$

We are done. □

B. $\kappa = \frac{\dot{x}\ddot{y} - \ddot{x}\dot{y}}{(\dot{x}^2 + \dot{y}^2)^{3/2}}$.

Proof. We can embed $\boldsymbol{\alpha}$ into $\mathbb{R}^3$ by constructing an equivalent $\boldsymbol{\alpha} : \mathbb{R} \rightarrow \mathbb{R}^3$, $\boldsymbol{\alpha} = (x(u), y(u), 0)$. We can see that $\dot{\boldsymbol{\alpha}} = (\dot{x}(u), \dot{y}(u), 0)$ and $\ddot{\boldsymbol{\alpha}} = (\ddot{x}(u), \ddot{y}(u), 0)$. We compute that

$$\dot{\boldsymbol{\alpha}} \times \ddot{\boldsymbol{\alpha}} = \dot{x}\ddot{y} - \ddot{x}\dot{y}.$$

We also compute that

$$|\dot{\boldsymbol{\alpha}}| = \sqrt{\dot{x}^2 + \dot{y}^2}.$$

Using the identity proven in (A), we see that

$$\kappa = \frac{\dot{\boldsymbol{\alpha}} \times \ddot{\boldsymbol{\alpha}}}{(|\dot{\boldsymbol{\alpha}}|)^3} = \frac{\dot{x}\ddot{y} - \ddot{x}\dot{y}}{(\dot{x}^2 + \dot{y}^2)^{3/2}}$$

Thus, we are done. □

C. $\frac{d\mathbf{T}}{ds} = \kappa \mathbf{N}$.

Proof. $\mathbf{T}$ exists only for a regular curve, and therefore we assume $\boldsymbol{\alpha}$ to be regular. By definition, $\mathbf{T} = \frac{\dot{\boldsymbol{\alpha}}}{|\dot{\boldsymbol{\alpha}}|}$. Because $\boldsymbol{\alpha}$ is regular there exists an arc length parameterization $s(u)$ that gives a $\mathbf{T} = \dot{\boldsymbol{\alpha}}(s)$, because $|\dot{\boldsymbol{\alpha}}(s)| = 1$ for all u . Differentiating with respect to s , we compute

$$\frac{d\mathbf{T}}{ds} = \ddot{\boldsymbol{\alpha}}(s).$$

We now show that $\ddot{\alpha}^\perp = \ddot{\alpha}$ for all u . Note that because α is arc length parameterized, $|\dot{\alpha}| = 1$ always, and thus $1 = \dot{\alpha} \cdot \dot{\alpha}$. We compute

$$\frac{d}{du}(1) = \frac{d}{du}(\dot{\alpha} \cdot \dot{\alpha}),$$

and so

$$0 = \ddot{\alpha} \cdot \dot{\alpha} + \dot{\alpha} \cdot \ddot{\alpha} = 2(\dot{\alpha} \cdot \ddot{\alpha}).$$

The above equation implies that $\dot{\alpha}$ is perpendicular to $\ddot{\alpha}$, since their dot product is 0. Therefore, there is no component of acceleration that points in the direction v ; as such, it must be the case that $\ddot{\alpha}^\perp = \ddot{\alpha}$. Note that $\kappa \mathbf{N} = \frac{|\ddot{\alpha}^\perp|}{|\dot{\alpha}|^2} \frac{\ddot{\alpha}^\perp}{|\ddot{\alpha}^\perp|}$. Since α is arc length parameterized, $|\dot{\alpha}| = 1$ always, so $\kappa \mathbf{N} = \ddot{\alpha}^\perp$, which we have shown to be equal to $\ddot{\alpha}$. Therefore, $\frac{d\mathbf{T}}{ds} = \kappa \mathbf{N}$, and we are done. $\square$

- D. By assumption α is regular, and therefore there exists an arc length parameterization (A.L.P) of α . Let α be A.L.P-ed. Furthermore, let

$$\beta(v) = \frac{1}{\kappa(u_0)}(\mathbf{N}(u_0))(1 - \cos(v)) + \frac{1}{\kappa(u_0)} \sin(v)\mathbf{T}(u_0) + \alpha(u_0).$$

Note that $\cos(v)$ and $\sin(v)$ are infinitely differentiable, and thus β is smooth. Furthermore, note that $\dot{\beta} = \frac{1}{\kappa(u_0)} \sin(v)\mathbf{N}(u_0) + \frac{1}{\kappa(u_0)} \cos(v)\mathbf{T}(u_0)$, which can never be the zero vector. Therefore, β is regular, and an A.L.P exists. Let β be A.L.P-ed. We show that β satisfies (1)-(4).

- (1) We compute

$$\beta(0) = \frac{1}{\kappa(u_0)}(-\cos(0) + 1) + 0 + \alpha(u_0) = \alpha(u_0).$$

- (2) We compute

$$\mathbf{T}_\beta(0) = \frac{\dot{\beta}(0)}{|\dot{\beta}(0)|} = \frac{\sin(0)\mathbf{N}(u_0) + \cos(0)\mathbf{T}(u_0)}{\frac{1}{\kappa(u_0)}\kappa(u_0)}.$$

Therefore, $\mathbf{T}_\beta(0) = \mathbf{T}(u_0)$.

- (3) We compute

$$\mathbf{N}_\beta(0) = \frac{\ddot{\beta}^\perp(0)}{|\ddot{\beta}^\perp(0)|} = \frac{\cos(0)\mathbf{N}(u_0) - \sin(0)\mathbf{T}(u_0)}{\frac{1}{\kappa(u_0)}\kappa(u_0)}.$$

Therefore, $\mathbf{N}_\beta(0) = \mathbf{N}(u_0)$. From (C), we know that because β is A.L.P-ed, $\ddot{\beta}^\perp = \ddot{\beta}$, and therefore the equality above holds.

- (4) We compute

$$\kappa_\beta(0) = \frac{|\ddot{\beta}(0)|}{|\dot{\beta}(0)|} = \frac{\frac{1}{\kappa(u_0)}}{(\frac{1}{\kappa(u_0)})^2} = \kappa(u_0).$$

Therefore, $\kappa_\beta = \kappa(u_0)$.